

How frequently do people touch shared surfaces in indoor public settings, in contacts per hour, measured by behavioural observation or video? Give separate rates for private/residential rooms and shared public spaces. Also: what total area of high-touch surface (handrails, door handles, tabletops, switches, buttons) exists per room, in square metres, for a hotel or cruise cabin versus a public lounge or dining room?

Surface touch rates range from ~2/h in offices to ~770/h in gyms; no paper reports high-touch surface area per room.

Video- and observation-based studies quantify surface touch frequencies across diverse indoor settings, but none of the 20 identified papers measure the total physical area of high-touch surfaces per room in square metres (Zhang et al., 2024; Zhuang et al., 2023; Jin et al., 2022).

Touch Rates in Private and Semi-Private Rooms

In a graduate student office, each student averaged **five touches per minute** (~300/h), but 95.7% involved private surfaces around the toucher and only 1.2% touched surfaces belonging to others (Zhang et al., 2018). Public-facility surfaces in the same office received fewer than **1 touch per hour** per student (Zhang et al., 2018).

In university dormitories, primary surfaces (desks, phones, keyboards, mice) were touched **10.4–25.4 times per hour** (Yuan et al., 2024). Korean adults observed in indoor daily life contacted environmental items at varying rates: phones 38.1/person/2h, furniture 23.2/person/2h, and computer accessories 12.5/person/2h (Oh et al., 2021). Contact density was highest for occasionally-shared items (Category II, 56.8) versus individually-used items (Category I, 35.9) and public-use items (Category III, 3.4) (Oh et al., 2021).

Touch Rates in Shared Public Spaces

Setting	All-Surface Touch Rate	Public-Surface Touch Rate	Method
Gym	770.3/h (Zhang et al., 2024)	517.5/h (Zhang et al., 2024)	Video + wearable devices
Airport self-service check-in	—	473.5/h (Zhuang et al., 2023)	Video
Restaurant (staff)	761.0/h (Jin et al., 2022)	545.4/h (T+R) (Jin et al., 2022)	Video
Restaurant (diners)	449.9/h (Jin et al., 2022)	42.8/h (T+R) (Jin et al., 2022)	Video
Airport manual check-in	414.9/h (Zhuang et al., 2023)	55.8/h (Zhuang et al., 2023)	Video

Setting	All-Surface Touch Rate	Public-Surface Touch Rate	Method
Hotel lobby	~21/h (627 touches/324 people) (Ackerley et al., 2023)	~21/h (Ackerley et al., 2023)	Direct observation
Hospital ward	93.1/h (all items) (Cheng et al., 2015)	13.6/h (bedside rails) (Cheng et al., 2015)	Direct observation
Graduate office	~300/h total (Zhang et al., 2018)	<1/h per student (Zhang et al., 2018)	Video

FIGURE 1 Surface touch frequencies across indoor settings measured by behavioural observation or video, showing total and public-surface contact rates per hour.

Gym attendees made **770.3 surface contacts per hour**, with 517.5 involving public surfaces — the highest rate recorded across all settings (Zhang et al., 2024). Airport passengers touched public surfaces most in self-service check-in areas at 473.5/h, where machine screens alone received 371.7 touches/h (Zhuang et al., 2023). Escalator handrails in airports were touched 214.6 times/h (Zhuang et al., 2023).

In a hotel lobby, 324 individuals produced 627 touches across 13 fomites over 30 hours, with the elevator button (32%) and front desk counter (22%) being most frequently contacted (Ackerley et al., 2025; Ackerley et al., 2023). Hospital wards saw 93.1 contact-episodes/h across all environmental items, with bedside rails (13.6/h) and bedside tables (12.3/h) topping the list (Cheng et al., 2015).

High-Touch Surface Area Per Room

No identified study quantifies the total physical area of high-touch surfaces (handrails, door handles, tabletops, switches, buttons) in square metres for hotel rooms, cruise cabins, public lounges, or dining rooms. Hotel cleanliness research compared guest-perceived contact frequency with ATP bioluminescence readings but did not measure surface area (Park et al., 2019). The hotel lobby observation study catalogued 13 distinct fomites by touch frequency but reported no dimensional measurements (Ackerley et al., 2025). Surface touch network studies in offices and airports enumerated surfaces and touch events without recording surface areas (Wang et al., 2021; Zhuang et al., 2023).

The absence of area-normalized data represents a gap: existing literature provides per-person or per-hour touch counts but not the denominator of available high-touch surface area per room that would enable contact-rate-per-square-metre comparisons between private cabins and public lounges.

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